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Hybrid Offline-First HMS: Project Estimation

2A. Resource Estimation for Project

Executive Overview

Project Constraints: 4 Student Engineers | **Timeline:** 14 Weeks (3.5 Months) | **Estimated Effort:** ~750 Person-Hours | **Total Budget:** ~$50 – $100 USD

This estimation outlines the human, hardware, software, and financial resources required to deliver the core framework of the **Hybrid Offline-First Hospital Management System (HMS)**. Designed to minimize software licensing fees and infrastructure overhead, the project relies entirely on open-source technologies and low-cost embedded hardware.

1. Human Resources & Role Allocation

The workload is distributed among four student developers based on domain specialization to prevent development bottlenecks and streamline integration:

Role	Core Responsibilities	Target Deliverables
Dev 1: Desktop Application Lead	Windows GUI frontend (.exe), local database schema, input validation, keyboard navigation.	PyQt/C# WPF App, SQLite local DB, zero-latency UI.
Dev 2: Embedded Systems & LAN Specialist	Embedded display client, local networking protocol (sockets/MQTT), live queue UI.	Raspberry Pi/ESP32 display firmware, local LAN broadcaster.
Dev 3: Cloud Backend & Sync Engineer	Remote REST API, background synchronization worker, sync conflict resolution handler.	FastAPI/Node.js backend, PostgreSQL cloud DB, auto-sync worker.

Dev 4: Testing & Review Engineer	Stress testing, Vulnerability Detection, Reviewing code practices, Maintaining Documentation.	pytest scripts, LaTeX documentation, YAML files.

2. Time & Effort Estimation

- **Project Duration:** 14 Weeks (~3.5 Months)
- **Individual Commitment:** ~16 – 18 hours / week per student
- **Team Capacity:** ~50 Person-Hours / week
- **Total Work Effort: ~700 – 756 Cumulative Person-Hours**

Effort Distribution Across Phases:

- **Phase 1: Local Foundation (4 Weeks / ~200 Hours):** Database schema, Receptionist & Doctor desktop GUI layouts, local CRUD operations.
- **Phase 2: Embedded Display & LAN Networking (4 Weeks / ~200 Hours):** Socket/MQTT communication, Embedded display interface, local LAN integration.
- **Phase 3: Cloud Sync Engine & Integration (4 Weeks / ~200 Hours):** Cloud REST API setup, background worker thread, offline-to-online sync & conflict resolution.
- **Phase 4: Testing, Hardening & Documentation (2 Weeks / ~100 Hours):** Network failure simulation, executable compilation, demo preparation.

3. Hardware, Software & Financial Infrastructure

A. Hardware Resources

- **Development Workstations:** 4x Existing Student Laptops (Windows / Linux / macOS) — **$0**
- **Embedded Display Unit:** 1x Raspberry Pi 4 (or ESP32 dev kit) + HDMI Display/Monitor — **~$45 – $70**
- **Testing Network:** 1x Standard Wi-Fi Router / Ethernet Switch (to test offline local LAN discovery) — **$0 (Existing)**

B. Software & Licensing Stack

- **Desktop & Local DB:** Python (PyQt) / C# (.NET Core) + SQLite — **$0 (Open-Source)**
- **Embedded Stack:** MicroPython / Python (Tkinter/Pygame) — **$0 (Open-Source)**
- **Backend & Cloud Database:** Python FastAPI / Node.js + PostgreSQL — **$0 (Open-Source)**
- **Development IDEs:** VS Code / Visual Studio Community Edition — **$0 (Free/Open-Source)**

C. Cloud Hosting & Deployment

- **Cloud Backend Hosting:** Supabase / Render / AWS Free Tier — **$0 – $10 / month**

4. Risk & Contingency Estimation

Identified Risk	Potential Impact	Resource Mitigation Strategy
Sync Conflict Engine Complexity	Sync logic delays project timeline.	Implement a deterministic Last-Write-Wins rule set; avoid multi-master sync branching.
Hardware Procurement Delays	Embedded hardware not immediately available.	Build a desktop-based "Display Simulator" in software to test local LAN packets early.
Academic Overlap (Exams/Assignments)	Temporary drop in developer availability.	2-week buffer built into Month 4; front-load core features into Months 1 & 2.

Conclusion

The resource estimation confirms that the Hybrid Offline-First HMS is feasible within the given academic constraints. With 4 student engineers, approximately 700–756 person-hours, and a low-cost open-source technology stack, the project can be developed within 14 weeks. Proper resource allocation and risk mitigation help ensure timely development and successful delivery of the MVP.